

T104 - BASIC ELECTRICAL AND ELECTRONICS ENGINEERING
2 MARKS
UNIT 5 - DIGITAL ELECTRONICS

Boolean algebra - Reduction of Boolean expressions - De-Morgan's theorem - Logic gates
Implementation of Boolean expressions - Flip flops - RS, JK, T and D. Combinational logic -
Half adder, Full adder and Subtractors. Sequential logic - Ripple counters and shift registers.

1. Draw the diagram of JK flip flop. (JAN 2014)

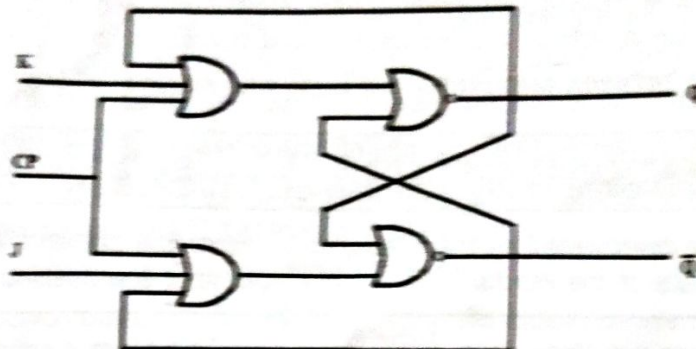


Fig. Logic circuit for Clocked JK Flip Flop

2. Write the truth table of half adder. (JAN 2014)

Table Truth table of Half adder

Input		Output	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

3. Draw the symbol for NAND gate and write its truth table. (MAY 2015)

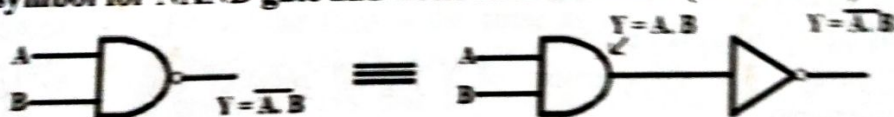


Fig. Standard Logic symbol of NAND Gate

Table 1 Truth table of NAND Gate

Input		Output
A	B	$Y = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

4. What is a flip flops? (MAY 2015)

Flip flops are synchronous bistable storage devices capable of storing one bit.

5. What are universal gates? (JAN 2016)

The NAND and NOR gates are called universal logic gates because all gates or logic circuits constructed using either NAND or NOR gate.

6. What is race around condition?(JAN 2016)

For J-K flip-flop, if $J=K=1$, and if $clk=1$ for a long period of time, then Q output will toggle as long as CLK is high, which makes the output of the flip-flop unstable or uncertain. This problem is called race around condition in J-K flip-flop.

7. Distinguish between combinational logic and sequential logic. (MAY 2016)

COMBINATIONAL CIRCUIT	SEQUENTIAL CIRCUIT
The output is discovered by the present state of the inputs.	Both the present input and past state output are used to identify the output.
Does not store data.	Can store a small amount of data.
Used in adders, encoders, multiplexer, etcetera.	Flip-flop and latches.
Circuits do not rely on the clock.	Clock is utilized for performing triggering functions.
No requirement of the feedback.	Feedback is required.

8. Define De-morgan's theorem. (NOV 2016)

De Morgan's First Theorem:

Statement - The complement of a logical sum equals the logical product of the complements.

Logic equation - $\overline{A+B} = \overline{A} \overline{B}$

De Morgan's Second Theorem:

Statement - The complement of a logical product equals the logical sum of the complements.

Logic equation - $\overline{A \overline{B}} = \overline{A} + \overline{B}$

9. What is a shift register? (NOV 2016)

A register which can shift its content either to the right flip-flop or to its left flip-flop is said to be shift register.

10. State De-morgan's theorem. (MAY 2017) (NOV 2016)

11. Write the truth table of half subtractor. (MAY 2017)

A	B	D	B ₀
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

12. State De-Morgan's theorem. (NOV 2017) (NOV 2016)
13. Write the truth table of half subtractor. (NOV 2017) (MAY 2017)
14. Distinguish between combinational logic and sequential logic. (MAY 2018) (MAY 2016)
15. Define De-morgan's theorem. (MAY 2018) (NOV 2016)
16. Which gates are called as universal gates? What are its advantages? (DEC 2018)
It is possible to use either only NAND gates or only NOR gates to implement any Boolean expression. This is so because a combination of NAND gates or a combination of NOR gates can be used to perform functions of any of the basic logic gates. It is for this reason that NAND and NOR gates are universal gates.
17. Name four different flip flops commonly available. (DEC 2018)
Commonly available flip-flops are JK flip-flop, RS flip-flop, D flip-flop and T flip-flop.
18. Name the universal gates. (MAY 2019) (JAN 2016)
19. What is the bottleneck in RS flip flop? (MAY 2019)
The fundamental disadvantage of the SR flip-flop is the indeterminate state of the output when the inputs S and R simultaneously assume the state of 1. A modification of the SR flip-flop, called the JK flip flop removes this problem.
20. State De Morgan's theorem. (NOV 2019) (NOV 2016)
21. What are basic properties of Boolean algebra? (NOV 2019)
Commutative: The commutative property says that binary operations AND and OR may be applied left to right or right to left. ($A \text{ AND } B$ is the same as $B \text{ AND } A$; $A \text{ OR } B$ is the same as $B \text{ OR } A$.)
Associative: The associative property says that given three Boolean variables, they may be ANDed or ORed right to left or left to right. ($(A \text{ AND } B) \text{ AND } C$ is the same as $A \text{ AND } (B \text{ AND } C)$; $(A \text{ OR } B) \text{ OR } C$ is the same as $A \text{ OR } (B \text{ OR } C)$.)
Distributive: The distributive property says that given three Boolean variables, the first AND the result of the second OR the third is the same as the first AND the second OR the first AND the third. ($A \text{ AND } (B \text{ OR } C) = (A \text{ AND } B) \text{ OR } (A \text{ AND } C)$). Also, the first OR the result of second AND the third is the same as the first OR the second AND the result of the first OR the third. ($A \text{ OR } (B \text{ AND } C) = (A \text{ OR } B) \text{ AND } (A \text{ OR } C)$.)
Identity: The identity property says that any value A AND the OR identity always returns A and that any value A OR the AND identity always returns A. ($A \text{ AND } 1 = A$; $A \text{ OR } 0 = A$.)
Complement: The complement property says that any value AND the compliment of that value equals the OR identity and that any value OR the compliment of that value equals the OR identity. ($A \text{ AND } (A') = 0$; $A \text{ OR } (A') = 1$.)
22. State Demorgan's theorem. (SEP 2020) (NOV 2016)
23. What are basic properties of Boolean algebra? (SEP 2020) (NOV 2019)

T104 - BASIC ELECTRICAL AND ELECTRONICS ENGINEERING**10 MARKS****UNIT - V DIGITAL ELECTRONICS**

Boolean algebra – Reduction of Boolean expressions - De-Morgan's theorem – Logic gates
 Implementation of Boolean expressions - Flip flops - RS, JK, T and D. Combinational logic -
 Half adder, Full adder and Subtractors. Sequential logic - Ripple counters and shift registers.

1. (a) Draw the diagram of Full adder and explain its operation. Write the truth table of Full adder. (5) (JAN 2014)

Full adder is a combinational circuit which performs the arithmetic addition of three input bits. The logical symbol for the full-adder is shown in fig(a) and the associated truth table is table

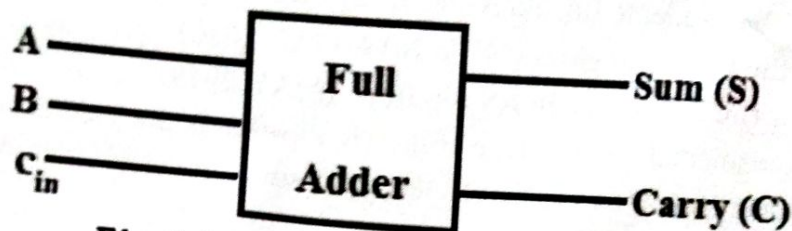


Fig. (a) Block diagram of Full Adder

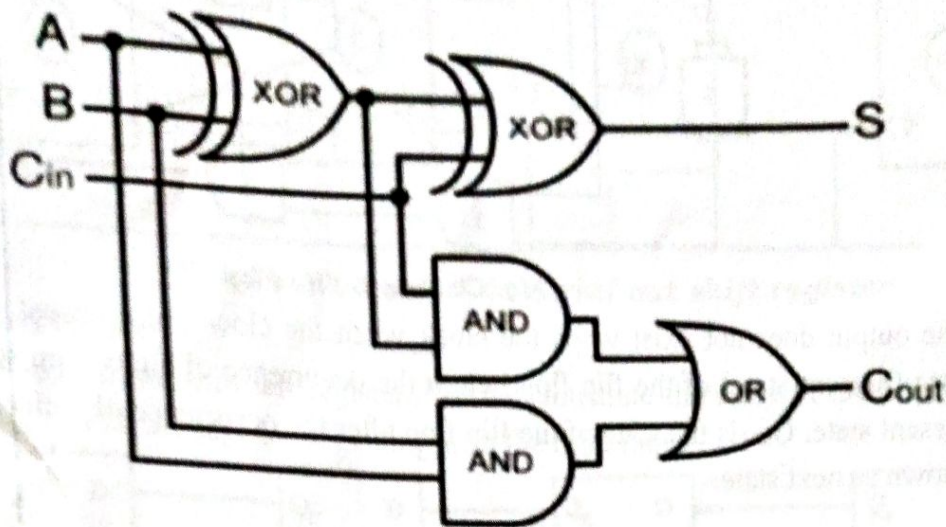
A	B	Carry-In	Sum	Carry-Out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

For the half-adder the sum of A and B is the XOR of those two variables. Similarly for the three variables A, B and C_{in} the sum becomes the XOR of all three inputs. From the truth table it is obvious that the three inputs are added and there are two outputs sum and carry. When any of the input values are '1' then the sum is '1' and carry is '0'. When any two of the inputs are '1' then the sum is '0' and carry is '1'. When all the three input values are '1' then both sum and carry is '1'.

The output column are combined and we get the following minimized expression for sum and carry

$$S = A \oplus B \oplus C_{in}$$

$$C = AB + C_{in}(A \oplus B)$$



(b) Draw the diagram of D flip-flop and explain its operation. Write the truth table of D flip-flop. (4) (JAN 2014)

D Flip Flop (Delay Flip Flop)

Looking at the truth table of SR flip flop we can realize that when both inputs are same the output either does not change or it is unpredictable (Inputs – 00, no change and inputs – 11 unpredictable). In many practical applications, these inputs conditions are not required. These input conditions can be avoided by making them complement to each other. This modified clocked SR flip flop is known as D flip-flop. As shown in fig, D input goes directly to the S input, and its complement is applied to the R input, through gate 5. Therefore, only two input conditions exist, either $S = 0$ and $R = 1$ and $S = 1$ and $R = 0$. When $D = 1$, $S = 1$ and $R = 0$ and then $D = 0$, $S = 0$ and $R = 1$. Therefore, during the occurrence of clock pulse if $D = 1$, the Q output is set and if $D = 0$, the output is reset. The truth table of clocked D flip flop is shown in Table.

Table Truth table of Clocked D flip flop

CP	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1

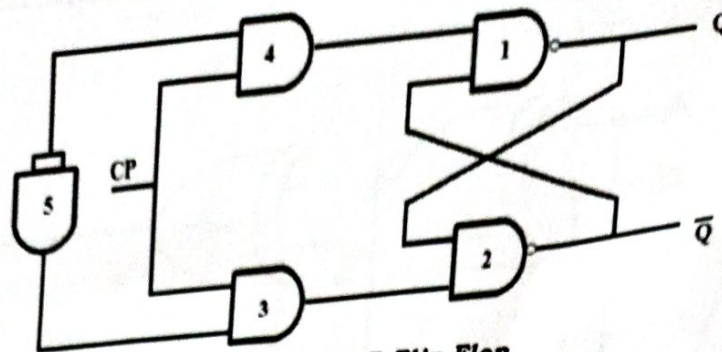


Fig. Clocked D Flip Flop

The output does not exist when the clock when the clock pulse absent. Q_n is the binary state (present state) of the flip flop before the occurrence of a clock pulse. It is known as present state. Q_{n+1} is the state of the flip flop after the occurrence of a clock pulse which is known as next state.

2. Write detailed notes about the operation of shift register with necessary diagram. (9) (JAN 2014)

A register that is capable of shifting data one bit at a time is called a shift register. The logical configuration of a serial shift register consists of a chain of flip-flops connected in cascade, with the output of one flip flop being connected to the input of its neighbor. The operation of the shift register is synchronous; thus each flip flop is connected to a common clock. Using D flip-flops forms the simplest type of shift-registers. The basic data movement possible within a four-bit shift register is shown in fig.

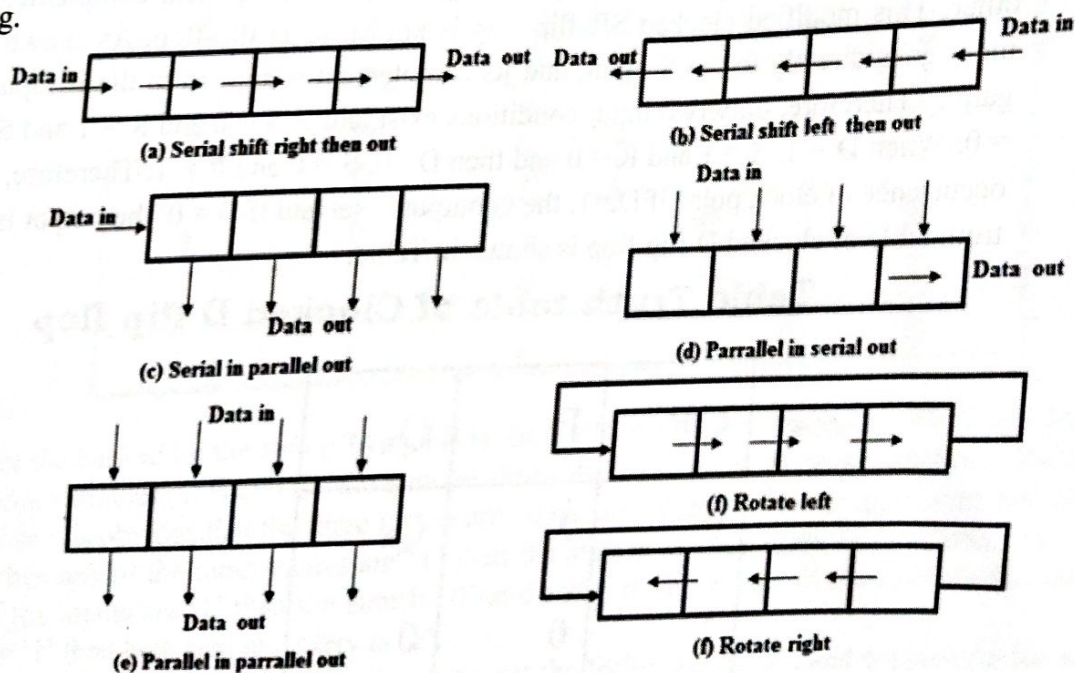


Fig. Data movement diagram

Figure shows the circuit diagram for a four-bit serial in-serial out shift register implemented using D flip-flops. Assuming that the register is initially clear

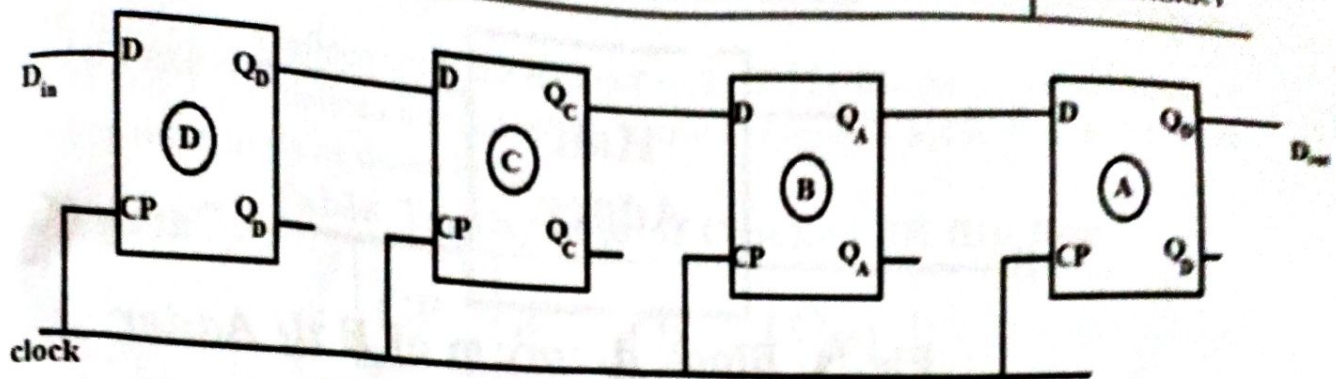


Fig. Four-bit serial in-serial out shift register

a) Shift left register

Figure below shows shift left register. We will illustrate the entry of the four bit binary number 1111 into the register beginning with the right most.

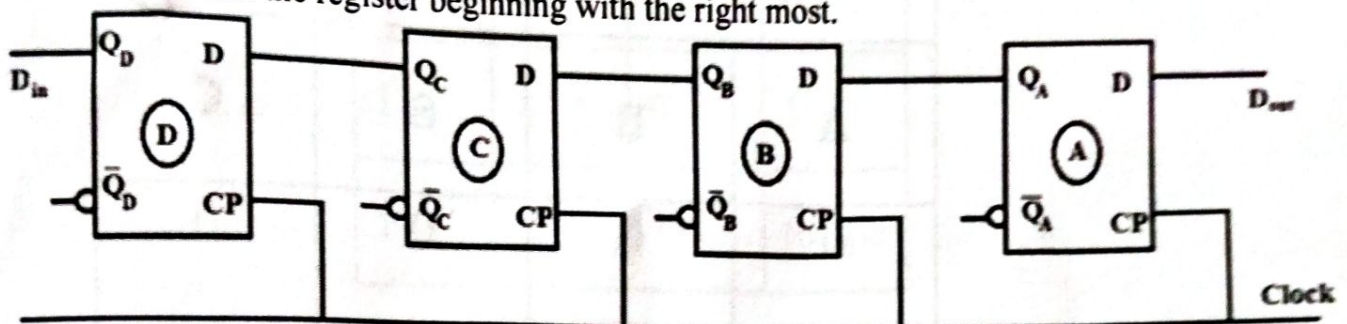


Fig. Logic Diagram for Shift Left Register

Case 1 Initially, register is cleared. So $Q_A Q_B Q_C Q_D = 0000$, When the data 1111 is applied serially. The left most 1 is applied to D_{in} , $D_{in} = 1$, $Q_D Q_C Q_B Q_A = 0000$. The arrival of the first falling clock edges sets the right most (D flip flop) and the stored information as follows $Q_D Q_C Q_B Q_A = 0001$

Case 2 When the next negative clock edges hits, the flip flop sets and the register contents $Q_D Q_C Q_B Q_A = 0011$

Case 3 The third negative clock edges, the register contents $Q_D Q_C Q_B Q_A = 0111$

Case 4 The fourth negative clock edges, the register contents $Q_D Q_C Q_B Q_A = 1111$

3. . (MAY 2014)
 4. . (JAN 2015)
 5. Explain the operation of full adder with neat diagram. (9) (MAY 2015) (JAN 2014)
 6. With neat diagram, explain the operation of shift register. (9) (MAY 2015) (JAN 2014)
 7. (a) Design a half adder. (3) (JAN 2016)
- Half adder is a logic circuit, which is used to perform two bit binary addition. Figure (a) shows the block diagram of half adder and the table shows the truth table of half adder.

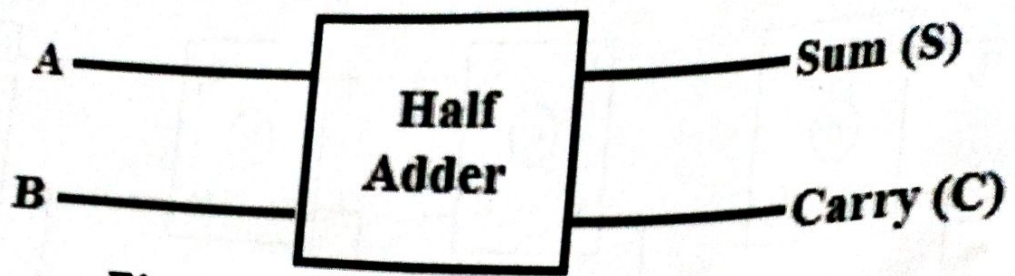


Fig.(a) Block diagram of Half Adder

Table Truth table of Half adder

Input		Output	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Observe from table that we may derive logical expressions for the Sum (S) and carry output (CO) bits. Namely, $CO = AB$, $S = A \oplus B$. From these two expressions, the implementation of the half adder function is as illustrated in fig (b).

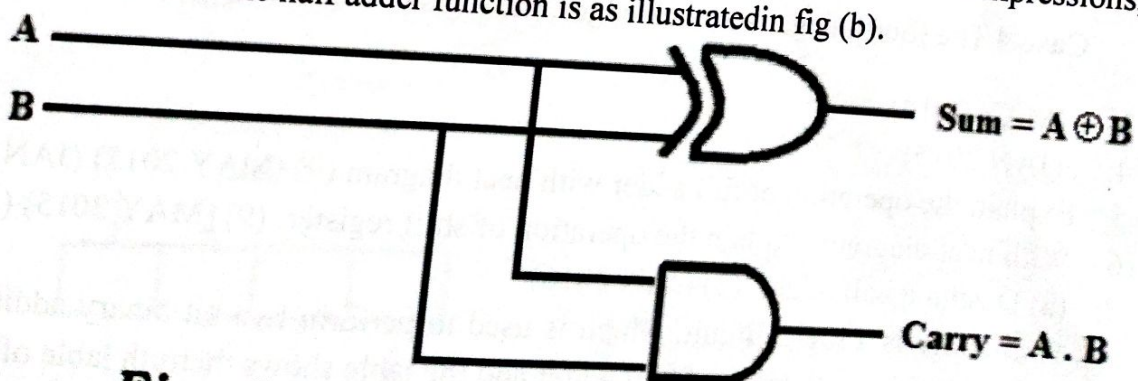


Fig. (b) Logic Circuit for Half Adder

(b) Explain the working of JK flip-flop. (6) (JAN 2016)

A JK flip-flop is a refinement of the RS flip-flop. In JK Flip-flop the unpredictable state in the RS flip-flop is defined inputs J and K behave like inputs S and R to set and reset the flip-flop respectively as shown in fig.

Table Truth table of Clocked JK flip flop

CP	J	K	Q_{n+1}
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	toggle

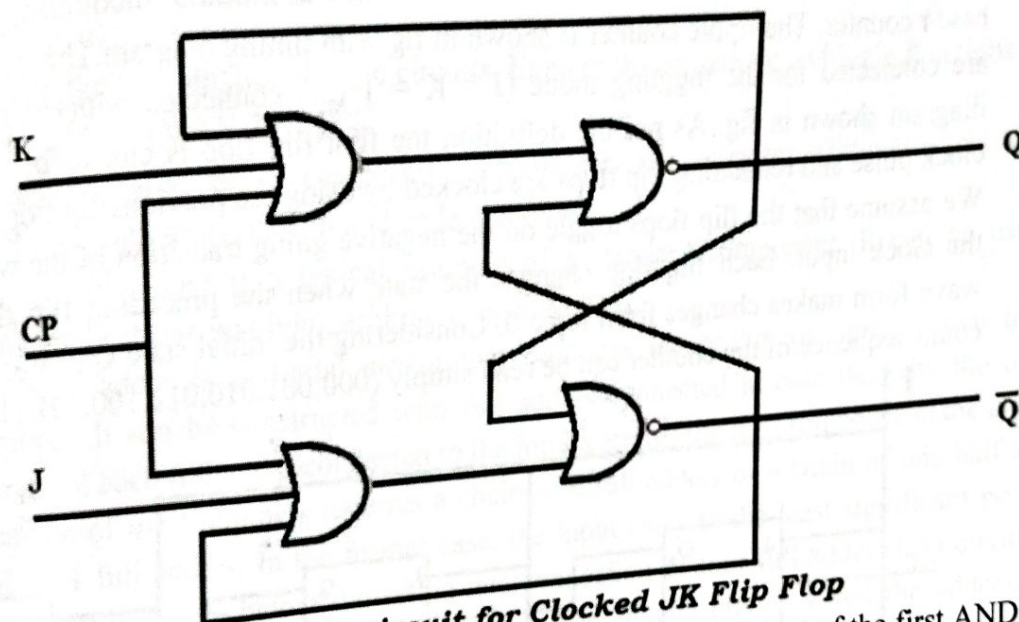


Fig: Logic circuit for Clocked JK Flip Flop

Case 1: when $J=0$, whatever be the value of Q'_n (0 or 1), the output of the first AND gate is 0. Similarly, when $K=0$, whatever be the value of Q_n (0 or 1), the output of the second AND gate is also 0. Therefore, when $J=0$ and $K=0$, the inputs to the basic flip-flop are $S=0$ and $R=0$. This condition forces the flip-flop into the same state.

Case 2: When $J=0$, $K=1$ and the previous state of flip flop is SET (i.e. $Q_n=1$, $Q'_n=0$), then $S=J * Q'_n=0 * 0=0$ and $R=K * Q_n=1 * 1=1$. Since $S=0$ and $R=1$, the flip-flop RESETS on the application of a clock pulse. But, if the flip-flop is already RESET (i.e., $Q_n=0$ and $Q'_n=1$), then the application of $J=0$ and $K=1$ will not alter the state of the flip-flop ($S=J * Q'_n=0 * 1=0$ and $R=K * Q_n=1 * 0=0$), and it remains in the RESET state.

Case 3: when $J = 1, K = 0$ and the previous state of the flip-flop is RESET ($Q_n = 0$ and $Q'_n = 1$), then $S = JQ'_n = 1 * 1 = 1$ and $R = KQ_n = 0 * 0 = 0$. Since $S = 1$ and $R = 0$ the flip-flop will SET on the application of a clock pulse. But if the flip flop is already set (i.e. $Q_n = 1, Q'_n = 0$) then the inputs $J = 1, K = 0$ will make $S = 0$ and $R = 0$, and the flip-flop remains in the SET state

Case 4: The condition $J = 1, K = 1$ deserves special attention. When $J = 1, K = 1$ and the previous state is a SET state (i.e. $Q_n = 1, Q'_n = 0$), then $S = JQ'_n = 1 * 0 = 0$ and $R = KQ_n = 1 * 1 = 1$. Since $S = 0$ and $R = 1$, the flip-flop RESETs on the application of a clock pulse, i.e. the flip-flop toggles from SET to RESET state. Again for $J = 1, K = 1$, if the previous state is a RESET state (i.e. $Q_n = 0, Q'_n = 1$), then $S = JQ'_n = 1 * 1 = 1$ and $R = KQ_n = 1 * 0 = 0$. Since $S = 1$ and $R = 0$, the flip-flop on the application of a clock pulse will toggle to the SET state. Hence, when $J = 1$ and $K = 1$, the flip-flop toggles on the application of the next clock pulse. Toggles means to switch to the opposite state. The flip-flop will continuously change its state when $J = K = 1$ and $CLK = HIGH$, resulting in an unstable output

8. Explain the operation of ripple counter. (9) (JAN 2016)

If starting from the initial state, the counter returns to the initial state after completed sequence of the input wave form, the counter is called as modulo (modulus) counter or a base r counter. The ripple counter is shown in fig with timing diagram. Three JK flip flops are connected for the toggling mode ($J = K = 1$ (or) connected with +5V) the timing diagram shown in fig. As per the definition the first flip flop is clocked by the external clock pulse and remaining flip flops are clocked by using the previous flip flop outputs. We assume that the flip flops toggle on the negative going transition of the waveform at the clock input. Each flip flop changes the state when the preceding flip flop or input wave form makes changes from logic 0. Considering the initial state $Q_A Q_B Q_C = 000$. The count sequence of the counter can be read simply (000, 001, 010, 011, 100, 101, 110, 111)

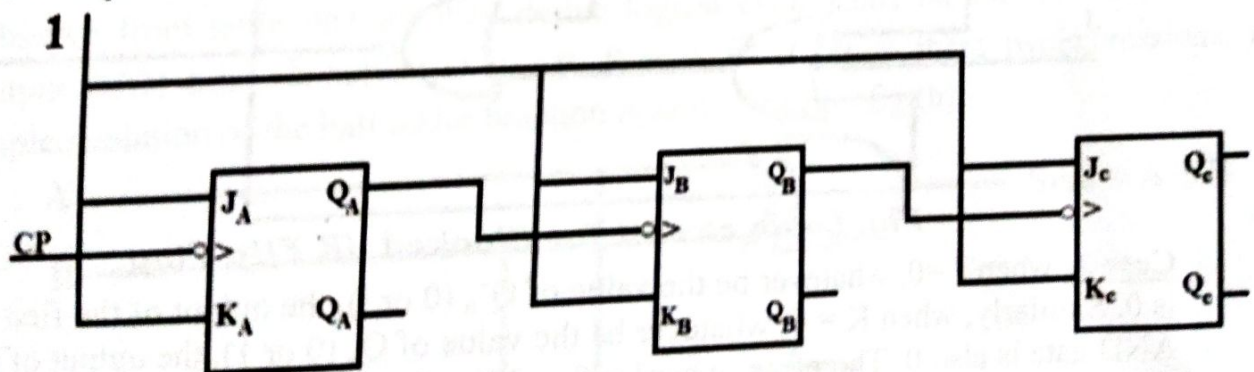


Fig Three bit Ripple Counter

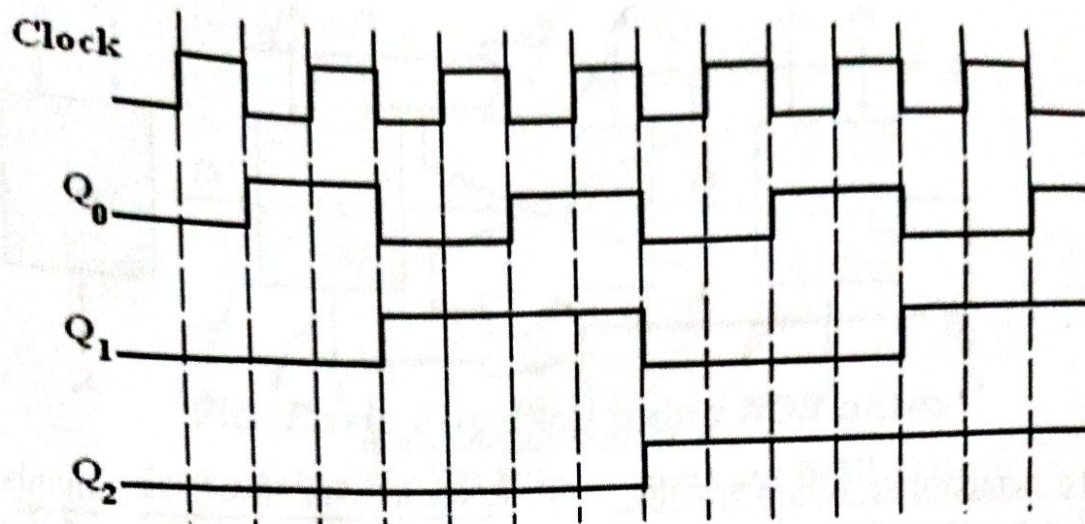


Fig. Timing diagram for 3 bit Ripple Counter

9. Explain in details the operation JK flip flop. (9) (MAY 2016) (JAN 2016)
10. Explain the operation of ripple counter. (9) (MAY 2016) (JAN 2016)
11. Discuss in detail about JK flip flop with logic diagram and truth table. (9) (NOV 2016) (JAN 2016)
12. Explain the working of half adder and full adder circuit. (9) (NOV 2016) (JAN 2014) (JAN 2016)
13. Classify the combinational logic circuits. Explain the arithmetic and logic functions using binary adders. (MAY 2017)

There are three main categories of combinational circuits: arithmetic or logical functions, data transmission and code converter.

The four-bit adder is a typical example of a standard component. It can be used in many applications involving arithmetic operations.

A binary adder is a digital circuit that produces the arithmetic sum of two binary numbers. It can be constructed with full adders connected in cascade, with the output carry from each full adder connected to the input carry of the next full adder in the chain. Addition of n -bit numbers requires a chain of n full adders or a chain of one-half adder and $n - 1$ full adders. In the former case, the input carry to the least significant position is fixed at 0. Figure below shows the interconnection of four full-adder (FA) circuits to provide a four-bit binary ripple carry adder. The augend bits of A and the addend bits of B are designated by subscript numbers from right to left, with subscript 0 denoting the least significant bit. The carries are connected in a chain through the full adders. The input carry to the adder is C_0 , and it ripples through the full adders to the output carry C_4 . The S outputs generate the required sum bits. An n -bit adder requires n full adders, with each output carry connected to the input carry of the next higher order full adder.

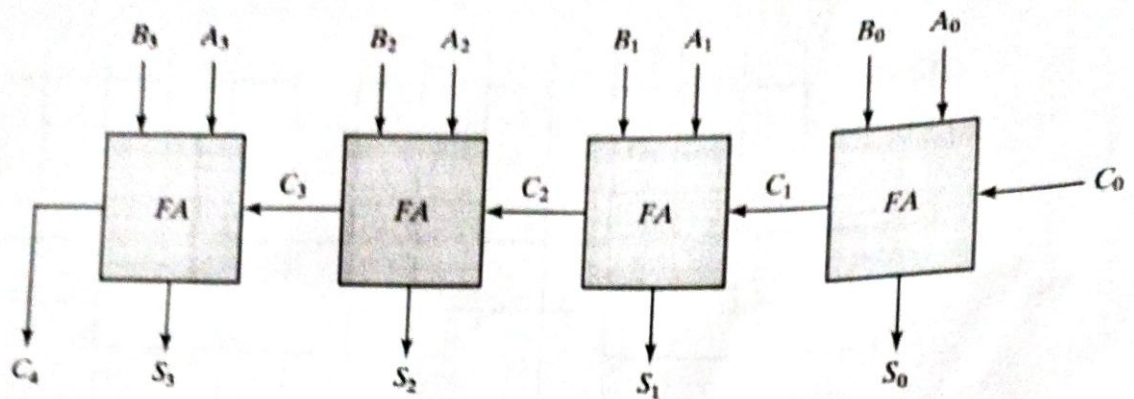


FIGURE Four-bit adder

To demonstrate with a specific example, consider the two binary numbers $A = 1011$ and $B = 0011$. Their sum $S = 1110$ is formed with the four-bit adder as follows:

Subscript i :	3	2	1	0	
Input carry	0	1	1	0	C_i
Augend	1	0	1	1	A_i
Addend	0	0	1	1	B_i
Sum	1	1	1	0	S_i
Output carry	0	0	1	1	C_{i+1}

The bits are added with full adders, starting from the least significant position (subscript 0), to form the sum bit and carry bit. The input carry C_0 in the least significant position must be 0. The value of C_{i+1} in a given significant position is the output carry of the full adder. This value is transferred into the input carry of the full adder that adds the bits one higher significant position to the left. The sum bits are thus generated starting from the rightmost position and are available as soon as the corresponding previous carry bit is generated. All the carries must be generated for the correct sum bits to appear at the outputs.

14. What do you understand by sequential logic circuit? Define and discuss about the basic SR flip-flop with its truth table. (MAY 2017)

The simplest type of flip flop is the Reset Set (RS) flip flop. It can be constructed using two NAND gates (or) two NOR gates. Figure below show the basic RS flip-flop using two NOR gates are cross coupled so that the output of NOR gate 1 is connected to one of the inputs of NOR gate 2 and vice versa.

The flip flop has two outputs Q and Q' , and two inputs, Set and Reset (S and R). the basic operation of flip flop as explained in the four cases as follows

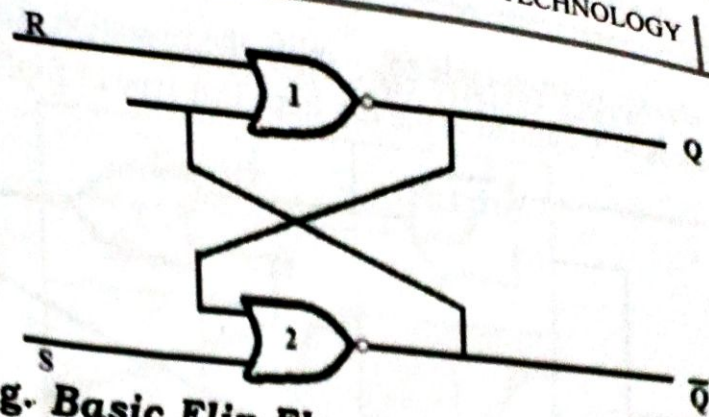


Fig. Basic Flip Flop using NOR gates
Truth table for RS Flip Flop using NOR Gate

R	S	Q	$\bar{Q}$	State
0	0	NC	NC	No change
0	1	1	0	Set
1	0	0	1	Reset
1	1	1/0	0/1	Indeterminate condition

Case 1 $R = 0$ and $S = 1$

In this case, the previous outputs are assumed as $Q = 1$. So the S input of NOR gate 2 is at logic 1, hence its output, Q' is at logic 0. Both inputs to NOR gate 1 are now at logic 0. So that its output, Q is at logic 1.

Case 2 $R = 1$ and $S = 0$

In this case, R input of NOR gate 1 is at logic 0, hence its output Q is at logic 1 when the previous outputs are considered as $Q = 0$ and $Q' = 1$. Both inputs of NOR gate 2 is logic 0. So that its output Q' is at logic 1.

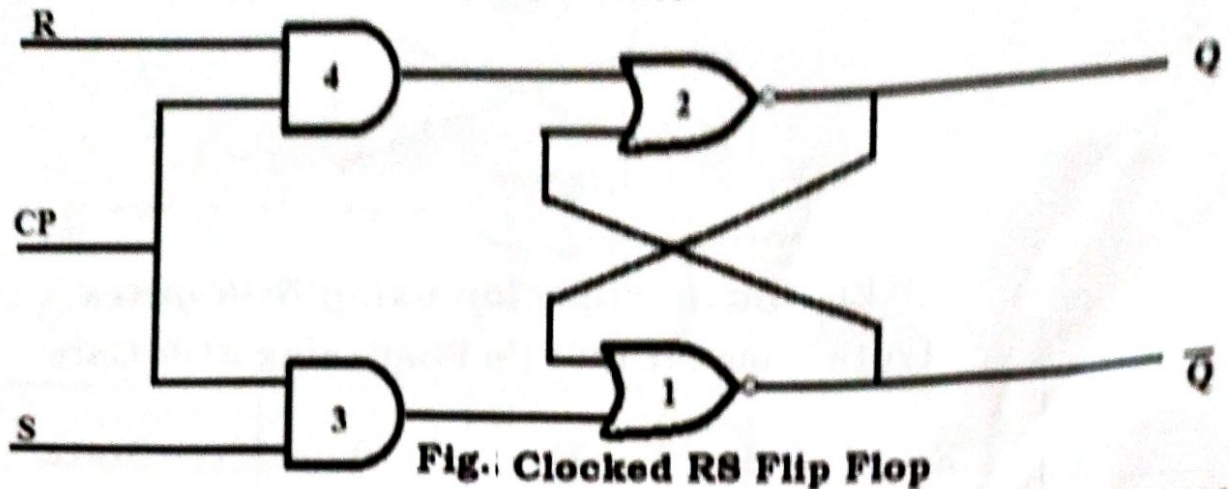
Case 3 $R = 0$ and $S = 0$

Let us assume that initially $Q = 0$ and $Q' = 1$. With $Q' = 0$ both inputs of NOR gate 1 are at logic 0. So its output Q is at logic 1. With $Q = 1$, one input of NOR gate 2 is at logic 1, hence its output Q' is at logic 0. This shows that when S and R both are low, output state does not change.

Case 4 $R = 1$ and $S = 1$

When R and S both are at logic 1, they force the outputs of both NOR gates to the low state ($Q = 0$ and $Q' = 0$). So we call this an indeterminate or prohibited state. This condition also violates the basic definition of a flip flop which requires Q' to be the complement of Q. Thus in normal operation this condition must be avoided by making sure that 1's are not applied to both inputs simultaneously. The truth table of basic flip flop using NOR

gates is shown in truth table to simplify the operation. To make synchronization of flip-flop, clock is also applied to the flip flop. This type of flip flop is called clocked flip flop.



15. (a) Draw the diagram of Full adder and explain its operation. Write the truth table of Full adder. (5) (NOV 2017) (JAN 2014)
- (b) Draw the diagram of D flip-flop and explain its operation. Write the truth table of D flip-flop. (5) (NOV 2017) (JAN 2014)
16. Write detailed notes about the operation of Ripple counter with necessary diagram. (9) (NOV 2017) (JAN 2016)
17. Explain in detail about digital to analog converter. (9) (MAY 2018) (Out of Syllabus)
18. Describe the operation of 4bit binary ripple counter. (9) (MAY 2018) (JAN 2016)
19. Draw and explain the circuit diagram of NAND, NOR, NOT and EX-OR gates with suitable truth table. (9) (DEC 2018)

NAND Gate: The NAND, which is composed of two (or) more inputs and a single output, is a very popular logic element because it may be used as a universal function. The term NAND is formed by the concatenation NOT-AND and implies an AND function with an inverted output. The standard symbol for the NAND gate is shown in figure below and its truth table is shown in Table 1. The logical operation of the NAND gate is such that the output is LOW (0) only when all the inputs are HIGH(1). NAND gate expression represent the function $Y = \overline{A \cdot B}$.

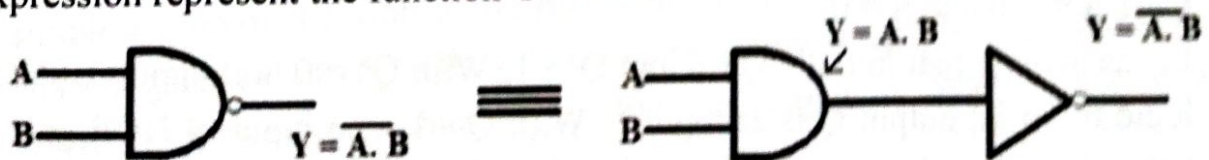


Fig.. Standard Logic symbol of NAND Gate

Table 1 Truth table of NAND Gate

Input		Output
A	B	$Y = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate: The NOR gate, which is composed of two (or) more inputs and a single output, also has a universal property. The term NOR is formed by the concatenation NOT-OR and implies an OR function with an inverted output. The standard symbol for the NOR gate is shown in Figure below and its truth table is shown in Table 2. The logical operation of the NOR is such that the output is HIGH (1) only when all the inputs are LOW (0). NOR gate expression represent the $Y = \overline{A + B}$.

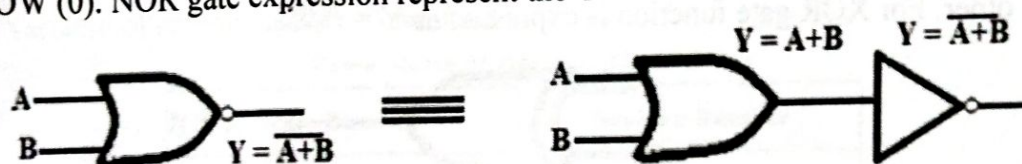


Fig. Standard logic symbol NOR Gate

Table 2 Truth table of NOR Gate

Input		Output
A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

NOT Gate: The inverter (NOT Gate) performs a basic logic function called inversion or complementation. The purpose of the inverter is to change one logic level (HIGH/LOW) to the opposite logic level. In terms of bits, it changes '1' to '0' and vice versa. The standard logic symbols for the inverter shown in Figure below. The truth table for the NOT gate is shown in Table3. NOT gate expression represent the function $Y = \bar{A}$.

Table 3 Truth table of NOT Gate

Input	Output
A	$Y = \bar{A}$
0	1
1	0

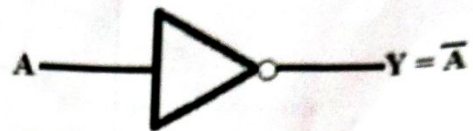


Fig. Standard logic symbol of NOT Gate

Exclusive-OR (XOR) Gate: This gate is usually formed from the combination of the other logic gates. However, because of its functional importance, this gate is treated as basic gate with its own unique symbols. The standard symbol and truth tables for the XOR gate is shown in Fig below and Table 4 respectively. The Exclusive -OR is an "inequality" function and the output is HIGH(1) when the inputs are not equal to each other. For XOR gate function is expressed as $Y = A \oplus B$.

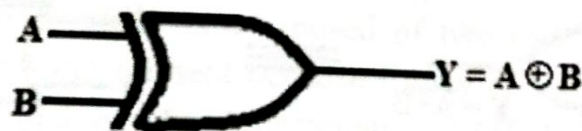


Fig. Standard logic symbol of XOR

Table 4 truth table for XOR Logic gate

Input		Output
A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

20. Explain the full adder circuit with its expression and truth table. (9) (DEC 2018) (JAN 2014)
21. With neat block diagram and waveforms, explain the operation of 4-bit ripple counter. (9) (MAY 2019) (JAN 2016)
22. Discuss about all the Boolean laws and prove them with simple example. (9) (MAY 2019)

LAWS OF BOOLEAN ALGEBRA

Commutative Law: The commutative law of addition for two variables is algebraically expressed as $A+B = B+A$. The commutative law of multiplication for two variables is expressed as $AB = BA$

Associative laws: The associative law of addition of three variables is expressed as $A + (B+C) = (A+B) + C$. The associative law of multiplication of three variables is expressed as $A(BC) = (AB)C$.

Distributive law: The distributive law of three variables is expressed as $A(B+C) = AB + AC$. This law states that ORing several variables and ANDing the result is equivalent of ANDing the single variable with each of the variables in the grouping, then ORing is the result.

RULES OF BOOLEAN ALGEBRA: By combining the laws of Boolean algebra, we form several useful rules that may be used in manipulating and simplifying Boolean algebra expressions. Table 5.8, note that in each case, A, B or C can either represent a single variable or a combination of variables.

Table Rules of Boolean Algebra

Rule Number	Boolean Number
1	$A+0=A$
2	$A+1=1$
3	$A.0=0$
4	$A.1=A$
5	$A+A=A$
6	$A+\bar{A}=1$
7	$A.A=A$
8	$A.\bar{A}=0$
9	$\bar{\bar{A}}=A$
10	$A+AB=A$
11	$A+B=B+A$
12	$(A+B)(A+C)=A+BC$